PERCONA

Ask Me Anything About MySQL 5.7 to 8 Upgrade

Vinicius Grippa

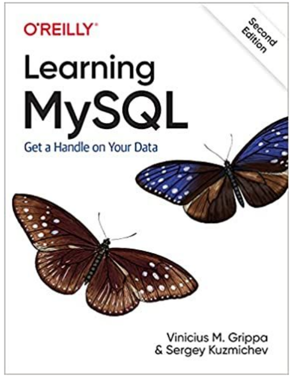
Lead Database Engineer

About me

Percona Database Engineer for 6 years

Working with databases for 18 years

Co-author of the book Learning MySQL



Who is Percona

For over 17 years Percona had been a leader in Open-Source databases, especially MySQL

Percona makes the lives of MySQL users easier by enabling developers to get to grips with their databases and automating management and operational tasks for database practitioners.

Our software and services for MySQL are trusted by thousands of enterprises to provide better performance and concurrency for their most demanding workloads, and delivers greater value to MySQL users with optimized performance, greater scalability and availability, enhanced backups, and increased visibility.

PERCONA

MySQL 5.7 @ EOL

Introduced October 2014

Best version of MySQL up to that time

MySQL 8.0

Introduced April 2018

The best version of MySQL

Many new features

MySQL 5.7 End Of Life

- What it really means
  - No more 5.7 development
- What if you do not upgrade?
  - No bug fixes, no updates
  - Lack of features
  - Huge technical debt
  - Very limiting

❏ Amazon Aurora 5.7 EOL 2024

Things To Note

- **Update issues**
  - Budget
  - Time
  - Rewrite code
    - GROUP BY example
    - Applications may need to be rewritten

- **Operation overhead**
  - Some queries slower
  - 4 byte UTF8MB4 takes more space than 1 byte Latin 1
  - Downtime
  - Update itself
  - Changes in configuration
  - Repartition tables

Your Options

- **Status Quo**
  - 5.7 Support from Percona

- **Upgrade to Oracle MySQL 8.0**
  - Community
  - Enterprise

- **Upgrade to Percona Server**
  - Enterprise features at Community Prices
  - With Percona for MySQL, you don't pay for optimized performance, greater scalability, enhanced backups, and increased visibility.
  - You use a production ready drop-in MySQL replacement for free and profit from more benefits compared to the standard MySQL database server
  - No software fee, and no vendor lock-in. Reduce the cost of MySQL database ownership with Percona!

Percona 5.7 Support

Yes, you can stay on 5.7 comfortably

Percona 5.7 EOL Support

MySQL 5.7 Post EOL Support subscription includes:

- Consultative support
- Operational support
- Break/fix support
- SLA commitment
- Releases for critical security issues and critical bug fixes

Software Covered in Percona's 5.7 EOL Support

- Covered Software
  - Percona Server for MySQL 5.7
  - Percona XtraDB Cluster 5.7
  - Percona XtraBackup 2.4

- EOL support offering is for Percona Server for MySQL 5.7, Percona XtraDB Cluster 5.7, and Percona XtraBackup only
  - Percona will release fixes into Open Source, so user can build their own binaries. Or they can migrate to Percona MySQL.
  - Same for EE users - EOL support is available if they migrate to Percona Server for MySQL 5.7.
  - Current and new customers
  - Percona Platform Standard or Premium Subscriptions Only
    - 1-year, 2-year, 3-year options available
  - Downloads provided:
    - On the platform(s) specified by the customer
    - Not available in the open public

PERCONA

Moving to 8.0 With Percona

The latest and greatest MySQL version

8.0 Options for Percona Customers

New customers

- Seamlessly migrate to Percona Distribution for MySQL 8.0

Existing customers

- Upgrade to version 8.0 without a hassle

What we promote:

- Migration support, upgrade support
- Post-migration database management (by Managed Services),
- 24/7 assistance with Percona support (by Support Services)
- Day 1 - Deployment
  - Install
  - set up
  - Configure
- Day 2 - Maintenance
  - upgrades, updates, patching
  - monitoring, observation
  - performance analysis and tuning

PERCONA

5.7 Options for Percona Customers

- **New customers**
  - Migrate to Percona Server for MySQL 5.7, Percona XtraDB Cluster 5.7, and XtraBackup 2.4

- **Existing customers**

  All 5.7. users are already eligible for EOL support.

- **We promote MySQL 5.7 Post-EOL Support (Percona Distro for MySQL 5.7)**:
    - Support for up to three years past the EOL date
    - Consultative and operational support
    - SLA commitment
    - Fixes for detected, reproducible critical bugs (CVE of Critical and High severity)
- Releases for predefined* platforms

PERCONA

Your Questions

There were previously submitted

We are running Percona ExtraDB Cluster 5.7 and had trouble upgrading in the past. Will it be easier to migrate from 5.7 to 8?
I would like to understand how hard (easy) in general the migration process is, and how such a migration should be planned and executed. Are ther any free tools that could help?
-Prerequisites Check - Things required to fix and things can be ignored for Upgrading from 5.8 to 8X
Are rumors true that 8.0 is slower than 5.x with latin1 schemas. Suggestions?
Concerned about 8 after seeing 8.0.33 has 140+ bug fixes. What is consensus opinion on stability of 8.
Why migration, why not upgrade?
200K schema's takes days for mysqlsh upgrade checker to complete. Will the upgrade alter the character set and collation during the upgrade procedure
Tuning in mysql8 for better performance
Is there any change?
Process and impact
Reserve word used in 5.7 needs to change prior to upgrade to MySQL8
Possible Performance issue.
Is that really a migration or upgrade? What things will break?
I am trying to replicate into a new 8.0 Percona Xtradb Cluster and no matter how I tune the server, there is still periods of replication lag. Is there anyway to tune the replication to keep up?

Concerned about 8 after seeing 8.0.33 has 140+ bug fixes. What is consensus opinion on stability of 8.

8.0 is very stable.

Most of those 140 bugs are minor, like updating library versions.

I would like to understand how hard (easy) in general the migration process is, and how such a migration should be planned and executed.
Are there any free tools that could help?

Use the upgrade checker in the MySQL Shell

mysqlsh> util.checkForServerUpgrade()

```
1) Usage of old temporal type
2) Usage of db objects with names conflicting with new reserved keywords
3) Usage of utf8mb3 charset
4) Table names in the mysql schema conflicting with new tables in 8.0
5) Partitioned tables using engines with non native partitioning
6) Foreign key constraint names longer than 64 characters
7) Usage of obsolete MAXDB sql mode flag
8) Usage of obsolete sql mode flags
9) ENUM/SET column definitions containing elements longer than 255 characters
10) Usage of partitioned tables in shared tablespaces
11) Circular directory references in tablespace data file paths
12) Usage of removed functions
13) Usage of removed GROUP BY ASC/DESC syntax
14) Removed system variables for error logging to the system log configuration
15) Removed system variables
16) System variables with new default values
17) Zero Date, Datetime, and Timestamp values
18) Schema inconsistencies resulting from file removal or corruption
19) Tables recognized by InnoDB that belong to a different engine
20) Issues reported by 'check table x for upgrade' command
21) New default authentication plugin considerations
```

Example of an error found by upgrade checker

2) Usage of db objects with names conflicting with reserved keywords in 8.0

Warning: The following objects have names that conflict with reserved keywords that are new to 8.0. Ensure queries sent by your applications use `quotes` when referring to them or they will result in errors.

More information: https://dev.mysql.com/doc/refman/en/keywords.html

davetest.reserved_word.over - Column name

Schema

Table

Column

Example of an error found by upgrade checker

2) Usage of db objects with names conflicting with reserved keywords in 8.0

Warning: The following objects have names that conflict with reserved keywords that are new to 8.0. Ensure queries sent by your applications use `quotes` when referring to them or they will result in errors.

More information:
https://dev.mysql.com/doc/refman/en/keywords.html

davetest.reserved_word.over - Column name

Schema

Table

Column

PERCONA

200K schema's takes days for mysqlsh upgrade checker to complete. Will the upgrade alter the character set and collation during the upgrade procedure?

You need to change the character set and/or collations.

200K - You may need to break this into smaller groups.

I am trying to replicate into a new 8.0 Percona Xtradb Cluster and no matter how I tune the server, there is still periods of replication lag. Is there anyway to tune the replication to keep up?

With any two two phase commit clustering system, there is always a slight lag.

Is there other traffic on the cluster network?

Is the lag consistent? Does it vary much with load?

Are you running long transactions?

PERCONA

Tuning in mysql8 for better performance?

Set the InnoDB buffer pool size to 75% of PHYSICAL RAM.

Run for a while, check hit rates.

Use Percona Monitoring and Management

We are running Percona ExtraDB Cluster 5.7 and had trouble upgrading in the past. Will it be easier to migrate from 5.7 to 8?

I would need to know more about your issues in the past.

https://forums.percona.com/

Are rumors true that 8.0 is slower than 5.x with latin1 schemas. Suggestions?

It can be.

MySQL 8.0 is optimized around UTF8MB4 which gives you the complete Unicode 9.0 characters.

Test and switch to UTF8MB4 if need be.

Your Questions

Submitted live

Wrap up

Submitted live

https://www.percona.com/navigating-mysql-5-7-end-of-life

https://www.percona.com/upgrading-to-mysql-8-0-with-percona

https://www.percona.com/post-mysql-5-7-eol-support

stacey.hunte@percona.com

Contact Information

Thank You!

David.Stokes@Percona.com
@Stoker